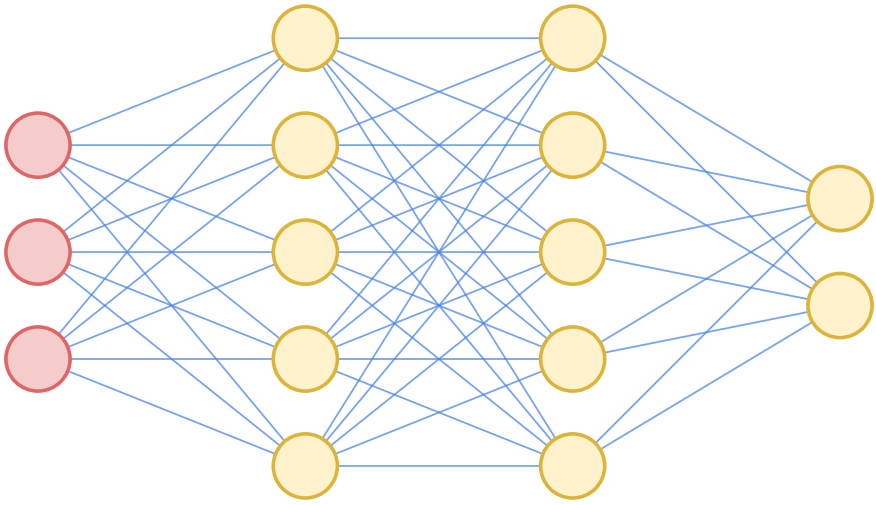


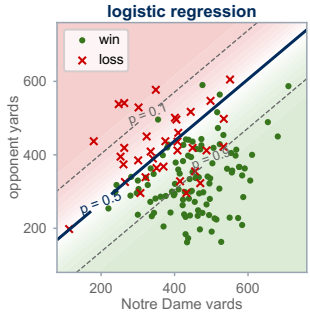
UNIT 02 · CHAPTER 1

Feed-Forward Networks

CSE 30124 · Introduction to Artificial Intelligence
William Theisen · University of Notre Dame

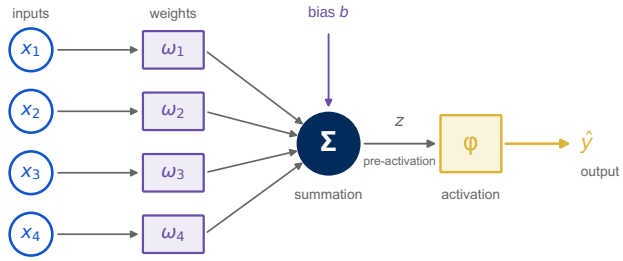
Predicting Game results

In **Model Evaluation** we asked "Did we win?" and a **logistic regression** answered with a straight line through the two teams' yards, getting `83.7%` of unseen games right.



Review: Perceptrons

Really, a **logistic regression** is the same thing as a **perceptron**, just with a _____ **activation function**. Generally, we can call this `unit` an _____



Deep learning is all about how we can use

_____

Artificial Neurons

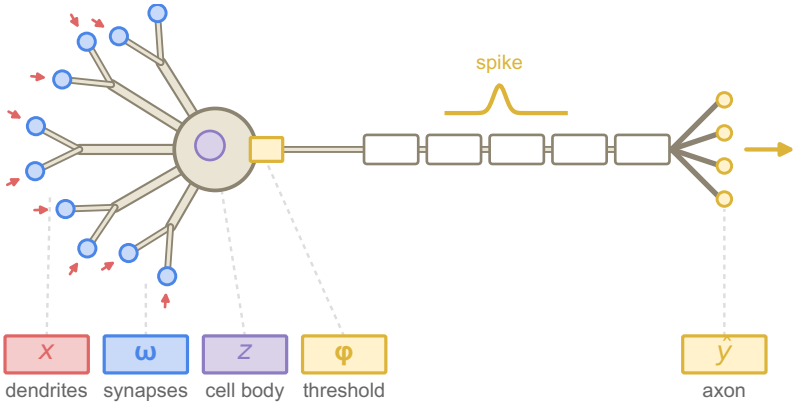
- _____ **(X):** The information the neuron is receiving
- _____ **(W):** defines how important a given input is for the neuron
- _____ **(b or ω_0):** A small adjustment to the weighted sum of the inputs
  - Think of this as shifting the neurons threshold to make it harder or easier to activate

- **Summation (Σ):** The weighted sum of the inputs and bias
  - represents the neurons "total input", also known as the **pre-activation**

The summation **transforms multiple inputs into one number** that summarizes the information that the neuron cares about.

- _____ **(ϕ):** A function applied to the pre-activation of the neuron
- _____ **($\hat{y}$):** The final resulting value from the activation function

Real Neurons

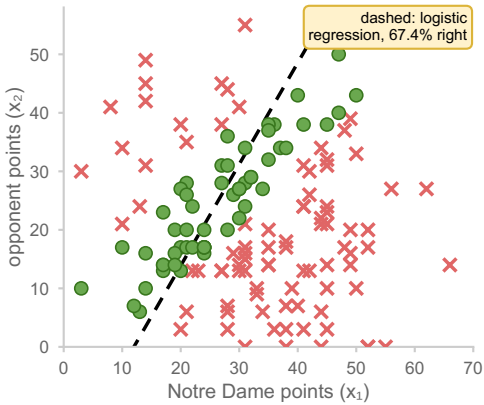


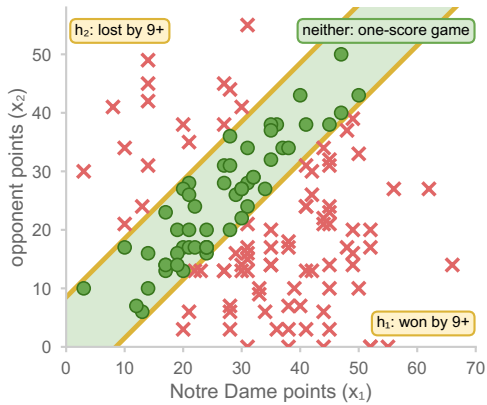
The **artificial neuron** is a (very) loose copy of a real one. A real neuron collects signals from other neurons on its **dendrites**, each synapse makes a signal count for more or less, and the neuron **fires a spike down its ____ only when the total crosses a threshold**.

Our brains have about `86` billion neurons and something like `100` trillion synapses. Our football network has `3` neurons.

The XOR Problem

What if instead of asking **win** or **loss**, we wanted to **decide whether a game was close or not (within 8 points)**

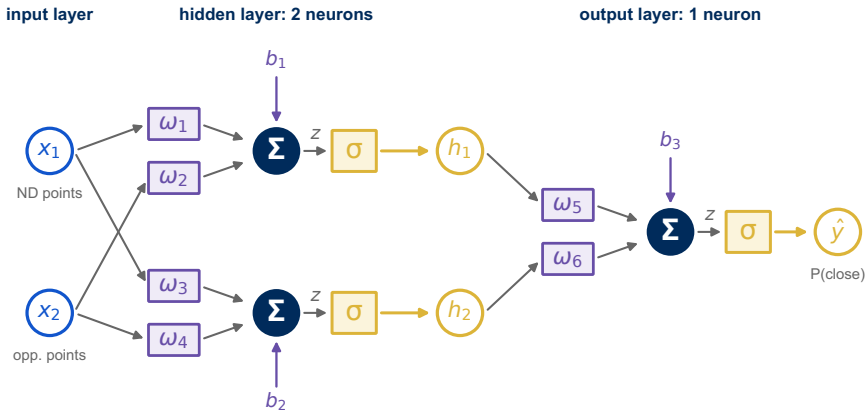




Asking if a game was close or not results in data that is very similar to the **XOR problem** we saw with **perceptrons**, where our **data is not**

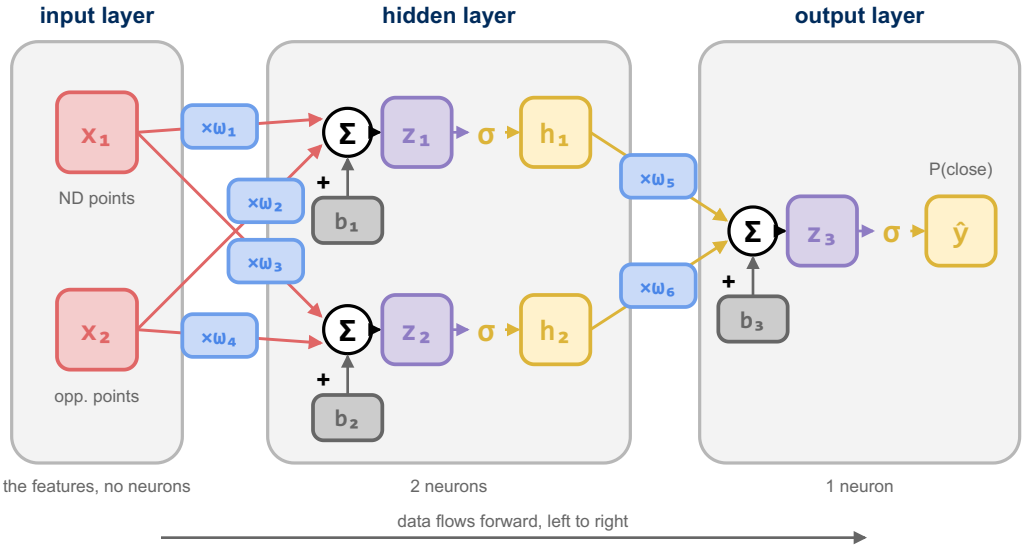
The XOR Problem

A **neuron** is capable of drawing a single line, but our data requires two lines to separate. We can have two **neurons** each create one of the two lines and use a _____ to **combine their output!**



Stacking **neurons** like this, so some take the outputs of others as their inputs, gives a **multi-layer perceptron (MLP)**, or a **feed-forward neural network (FFN)**

Neural Networks: Diagram

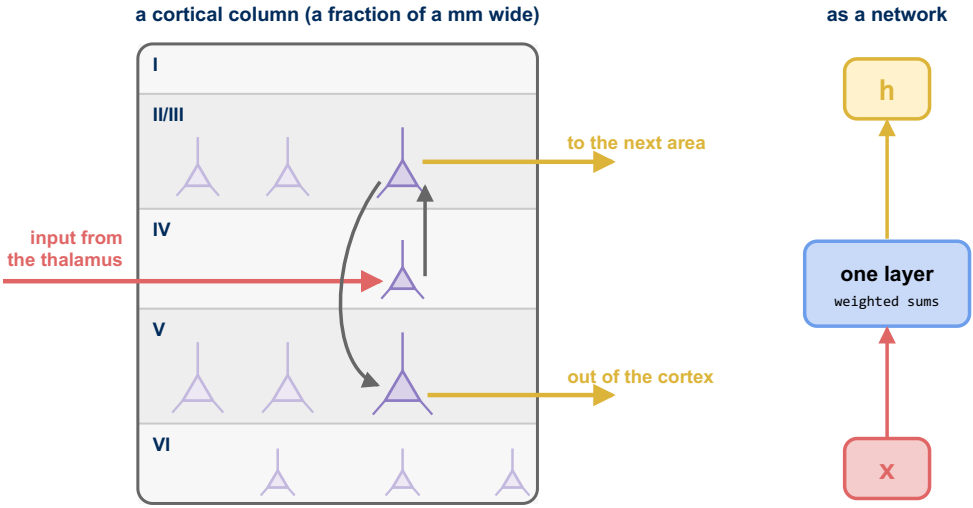


Combining our three **artificial neurons** gives us a simple **FFN**

The organization of neurons in a neural network is called its _____.

The Cortical Column

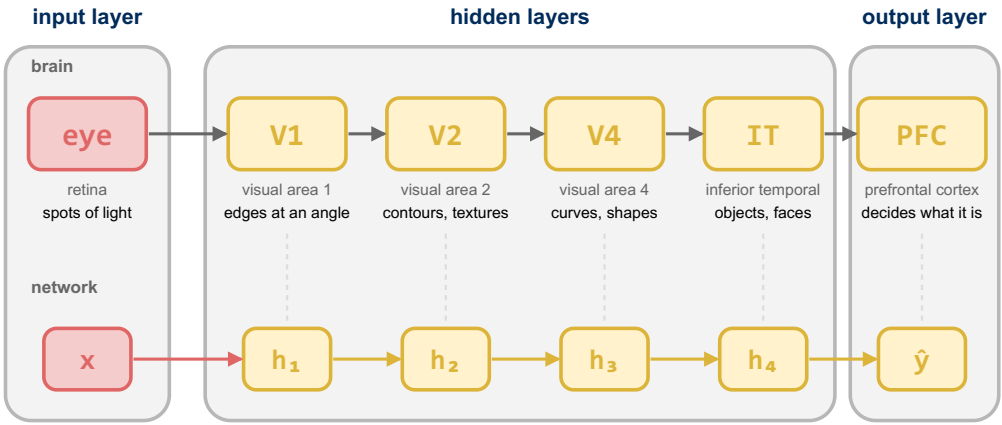
In **1957** Vernon Mountcastle found that neurons stacked straight down through a cat's cortex all responded to **the same patch of skin**. The cortex is built out of these _____, each a fraction of a millimeter wide.



Douglas and Martin (**1989**) proposed that columns all over the cortex run **the** _____, a bit like a network built by stacking the same kind of layer.

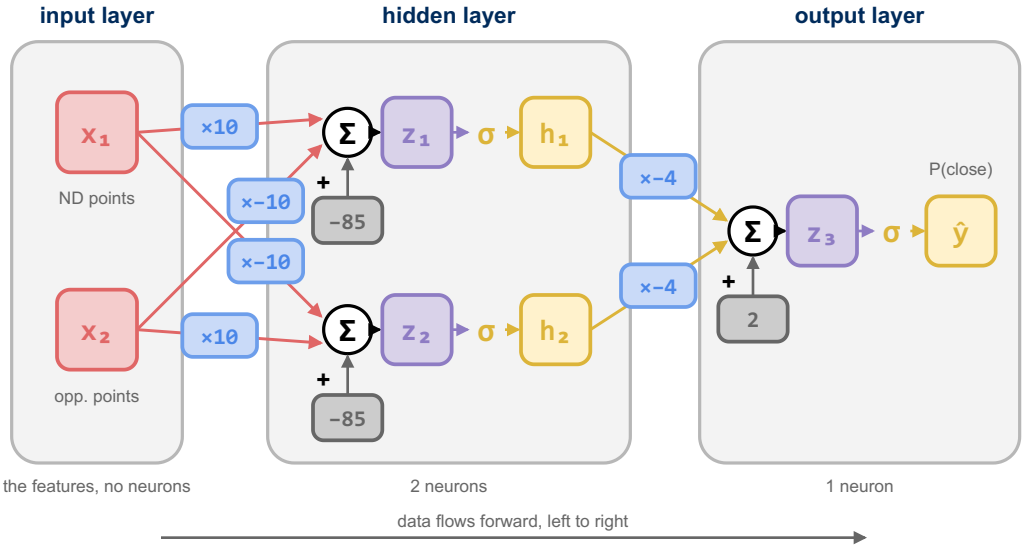
Depth in the Brain

Our network has an input layer, an output layer and **one layer** of neurons in between. Vision in the brain has the same shape, only _____: a chain of areas, each finding **more complex patterns** than the one before.



Felleman and Van Essen (1991) counted 32 visual areas in the monkey brain, arranged in about 10 levels.

Network Diagram with Weights



[0.1] two neurons look at the score

$$h_1 = \sigma(10x_1 - 10x_2 - 85) \qquad h_2 = \sigma(10x_2 - 10x_1 - 85)$$

[0.2] a third combines them

$$\hat{y} = \sigma(2 - 4h_1 - 4h_2)$$

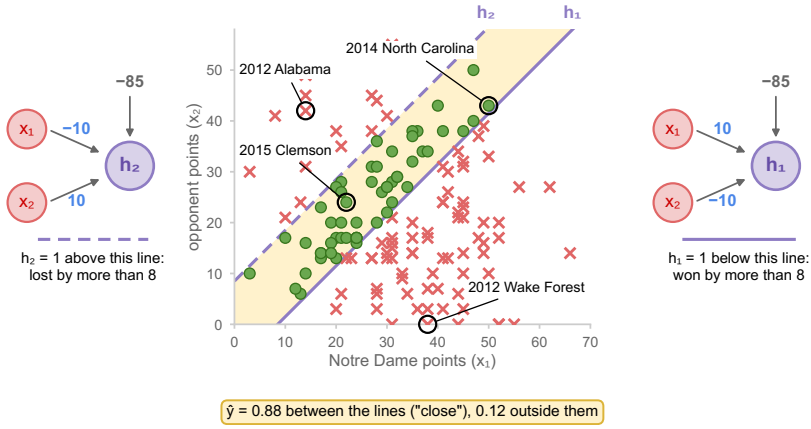
Producing Output from a Neural Network

Producing output with a **neural network** is called "performing the _____"

game	score	h_1	h_2	$\hat{y}$	close?
2014 North Carolina	50–43	0	0	0.88	yes
2015 Clemson	22–24	0	0	0.88	yes
2012 Wake Forest	38–0	1	0	0.12	no
2012 Alabama (title game)	14–42	0	1	0.12	no

Decision Regions

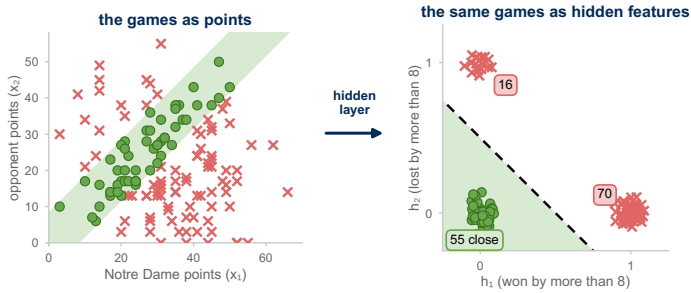
Our first two **neurons** each draw a _____, our third **neuron** combines the two boundaries to create a **decision region**



On all `141` games it gets every one right (`100%`), while no straight line could do better than `88.7%`.

Hidden Layer Features

Layers that don't produce the final output are called _____.



These are what transform our data into _____ points!

They create new features for our data that our third **neuron** can separate.

Hidden Features in the Brain

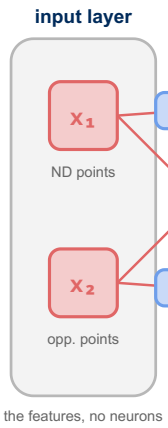
Our hidden layer turned each game into new features, **won by more than 8** and **lost by more than 8**. Do the hidden layers of a bigger network find features like the brain's? Yamins et al. (`2014`) trained a network to recognize objects in photos, then compared its hidden layers with neurons recorded in monkeys looking at the same photos.

- Its **middle hidden layers** matched ___, the area that responds to curves and shapes
- Its **last hidden layer** matched __, the area that responds to whole objects and faces
- Networks that were **better at recognizing objects** matched the brain better

The networks were never shown any brain data. Learning to do the same job, they ended up with **hidden features** a lot like the brain's.

Condensing the Hidden Layer

We can compactly represent the operations of the hidden layer as a single matrix multiplication.



X inputs, one row per game
<4, 2>

W_1 weights, o
<2, 2>

	x_1	x_2
2014 North Carolina	50	43
2015 Clemson	22	24
2012 Wake Forest	38	0
2012 Alabama	14	42

neuron 1

neuron 2

Weights as Matrices

four games times both hidden neurons' weights, plus the bias (added to every row)

$$Z_1 = XW_1^T + b_1 = \begin{bmatrix} 50 & 43 \\ 22 & 24 \\ 38 & 0 \\ 14 & 42 \end{bmatrix} \begin{bmatrix} 10 & -10 \\ -10 & 10 \end{bmatrix} + \begin{bmatrix} -85 & -85 \end{bmatrix}$$

each row is one game, each column one hidden neuron

$$= \begin{bmatrix} 70 & -70 \\ -20 & 20 \\ 380 & -380 \\ -280 & 280 \end{bmatrix} + \begin{bmatrix} -85 & -85 \end{bmatrix} = \begin{bmatrix} -15 & -155 \\ -105 & -65 \\ 295 & -465 \\ -365 & 195 \end{bmatrix}$$

the sigmoid of every cell, the A1 the code prints

$$A_1 = \sigma(Z_1) = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$$

NOTE: This example is a little goofy because the result of the transpose is the same matrix

Forward Pass of a Neural Network

```python
import torch

X = torch.tensor([[50., 43.], [22., 24.], [38., 0.], [14., 42.]])

W1 = torch.tensor([[10., -10.], [-10., 10.]])
b1 = torch.tensor([-85., -85.])

A1 = torch.sigmoid(X @ W1.T + b1)  # b1 is added to every row

print(X.shape, W1.shape, b1.shape)
print(A1.round(decimals=3))
```

OUTPUT

```
torch.Size([4, 2]) torch.Size([2, 2]) torch.Size([2])
tensor([[0., 0.],
        [0., 0.],
        [1., 0.],
        [0., 1.]])
```

similar to `sklearn` for machine learning, `PyTorch` is the library we will use to implement all of our **neural networks**.

The `@` symbol is how we multiply matrices and `.T` gives us the transpose of a matrix

A Forward Pass in the Brain

Thorpe, Fize and Marlot (1996) flashed photos for just 20 ms and asked people **"is there an animal in it?"**

Their brain signals for "animal" and "no animal" had already split apart about 150 ms after the photo appeared.

Neurons are slow (milliseconds per step), so 150 ms is only enough time for the signal to go up the visual hierarchy about _____. A first look at a scene seems to work **much like a** _____: no loops, no second guesses.

Activation Functions

Critically, **activation functions** introduce _____ into the model. This is what allows **neural networks** to learn complex patterns (beyond just linear relationships).

two layers with no activation function

$$Z_2 = (XW_1^T + b_1)W_2^T + b_2$$

multiply out

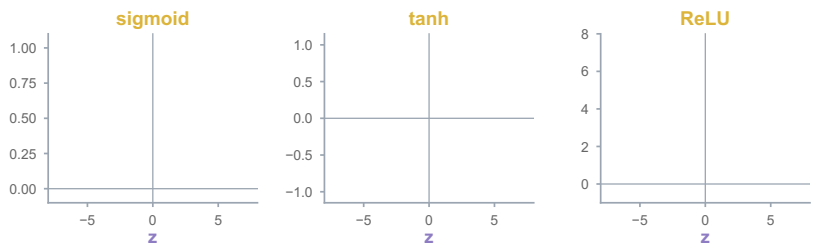
$$= X(W_2W_1)^T + (b_1W_2^T + b_2)$$

a single layer: its weights are the product of the two, its bias the bracket

$$= XW^T + b$$

If we didn't have **activation functions**, our **neural network** would be no more powerful than just a single **neuron**. However, this power comes at the cost of _____.

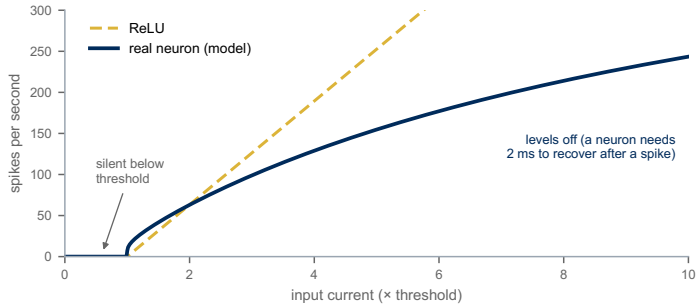
For our simple **neural network**, we are using the **sigmoid function** as the **activation function**, but it's not the only choice.



Other common options are the **hyperbolic tangent** and the **ReLU** function

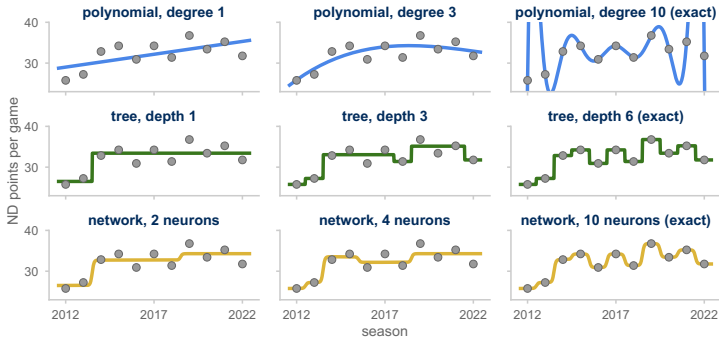
Activation Functions in Real Neurons

A real neuron stays **silent below its threshold**, then fires faster the harder it's pushed, until it levels off.



Near the threshold that's a lot closer to a _____ than to a sigmoid. Glorot et al. (2011) used this to argue for ReLU, along with the fact that only about 1–4% of the brain's neurons are active at once (**sparse**, like a ReLU layer's zeros).

Universal Approximation Theorem



The **universal approximation theorem** says a **neural network** with a single **hidden layer**, given enough **hidden neurons**, can approximate any continuous function

One Neuron, One Network?

Our artificial neuron adds up its inputs once. A real neuron's dendrites branch, and **each branch does its own non-linear sum** before the cell body adds them up.

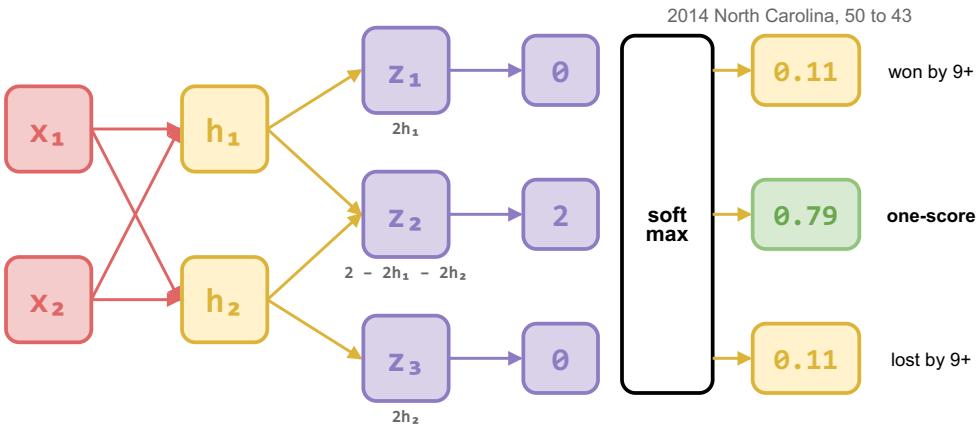
Poirazi, Brannon and Mel (2003) modeled a pyramidal neuron this way and found it behaves like a _____ **neural network**, every branch a hidden neuron.

Beniaguev, Segev and London (2021) trained networks to copy a detailed model of one cortical neuron, spike for spike. It took a network 7 **layers deep**.

So **one real neuron is already a small network**, and our 3-neuron FFN is a long way from even one of them.

Multiclass Classification

Another powerful new ability we gain with **neural networks**, is we are no longer limited to binary outputs, we can use them naturally for **multi-class classification**



Multi-class Classification

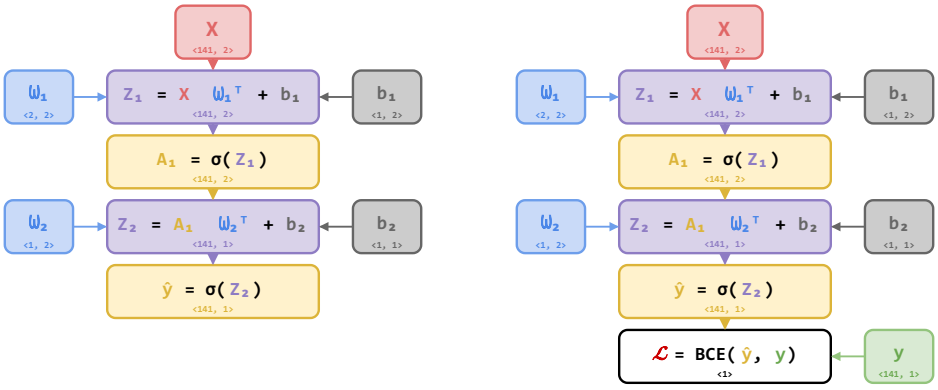
Our three output **scores** aren't **probabilities**. To turn them into probabilities that add up to 1, we use the _____ function:

$$\text{softmax}(\mathbf{z})_i = \frac{e^{z_i}}{\sum_{j=1}^{K} e^{z_j}}$$

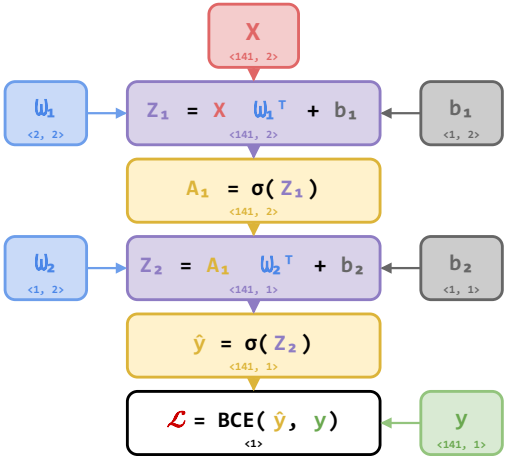
Softmax is the usual **activation function** for the _____ of a **classifier**. With only two classes it's the same as the _____.

Representing Neural Networks

Another common way to represent the **architecture** of a **neural network** is as a _____.

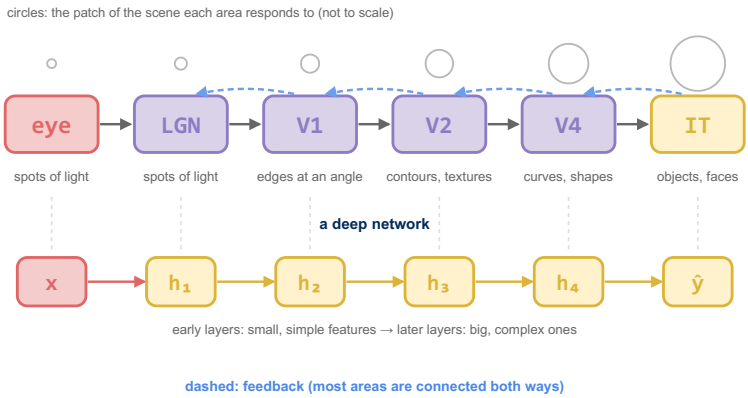


From the **computation graph**, we can also represent our **neural network** as a single large _____.



This is **critical for training** our **neural networks** (next class)

What FFNs Leave Out



- _____: most areas are connected both ways, and only about 5–10% of the input to the LGN comes from the eye (most of the rest comes back down from the cortex)
- **Spikes**: real neurons send all-or-nothing spikes, and their timing matters, not just one number
- **Learning**: nobody has found the brain doing what we'll do next lecture (**backprop**)

An FFN is a model of the brain's **first, fast sweep**, not the whole thing.

An Old Exam Question

```python
import torch
import torch.nn as nn
import torch.nn.functional as F

class SimpleNetwork(nn.Module):
    def __init__(self):
        super(SimpleNetwork, self).__init__()
        self.fc1 = nn.Linear(2, 32)
        self.fc2 = nn.Linear(32, 16)
        self.fc3 = nn.Linear(16, 1)

    def forward(self, x):
        x = F.relu(self.fc1(x))
        x = F.relu(self.fc2(x))
        x = torch.sigmoid(self.fc3(x))
        return x

criterion = nn.BCELoss()
```

a.) Draw the computation graph for the network. (4 pts.)

b.) Give the loss of the neural network (L) as a single statement of nested functions. (1 pt.)